

FreelanceHub - Full-stack freelancing marketplace platform.

INTRODUCTION:

Welcome to FreelanceHub — where connecting with the right talent or the right project is fast, transparent, and secure. Whether you're a client looking to hire or a freelancer looking for work, our platform offers everything you need to get things done.

Post a job or browse open listings across categories like Web Development, UI/UX Design, AI & Machine Learning, Data Science, and Digital Marketing. Every job includes clear budgets, required skills, and estimated timelines to help both sides make informed decisions.

With a user-friendly interface, clients can track proposals, hire freelancers, and manage active projects, while freelancers can build a portfolio, track earnings, and grow their reputation through verified reviews. Work from anywhere — on your phone, tablet, or computer.

Join our growing community of clients and freelancers to build meaningful working relationships. Start your journey today and open the door to new opportunities with FreelanceHub.

SCENARIO Based Case Study:

Background

"FreelanceHub" is an interactive freelancing marketplace designed to make hiring and getting hired simple, transparent, and effective. With a growing base of clients and freelancers, FreelanceHub supports projects across multiple categories, including Web Development, Design, Data Science, and Marketing. The platform focuses on combining smart job matching, transparent bidding, and built-in collaboration tools to help both sides get to a successful outcome faster.

Scenario

User Persona:

Name: Rohan Mehta

Age: 29

Profession: Independent Frontend Developer

Goal: Find consistent freelance web development projects and build a reliable client base.

Challenges: Unpredictable income, difficulty finding trustworthy clients, no clear way to showcase verified work history.

The Challenge

Rohan had tried reaching out to clients through personal networks and social media before, but found the process slow, unreliable, and hard to scale. He needed a platform that could:

- Surface relevant job opportunities matching his skill set
- Let him submit proposals with clear terms (bid, timeline, cover letter)
- Provide a secure way to communicate and deliver work with clients
- Build a visible, verified reputation through ratings and reviews

Solution: How FreelanceHub Helped

When Rohan registered on FreelanceHub, he set up a freelancer profile listing his skills, hourly rate, and a short bio. From there, the platform helped him with:

- Smart Job Matching: Filterable job listings by skill, category, and budget, so he only sees relevant work.
- Proposal & Hiring Flow: A structured way to submit bids and get hired directly on the platform, with hiring status tracked automatically.

Within two months, Rohan completed three projects on FreelanceHub, built up a 4.8 average rating, and secured a repeat client for ongoing work.

SYSTEM REQUIREMENTS:

To ensure smooth development, deployment, and usage of the FreelanceHub Web-Application, certain system prerequisites must be met. These requirements are categorized into software

and hardware specifications, all of which contribute to building a robust and scalable freelancing platform.

1. Software Requirements

These are the essential tools and platforms required to develop, test, and run the application efficiently.

- Operating System: Windows 10/11, macOS, or Linux — Supports cross-platform development and testing.
- Node.js (v16 or above): Provides the runtime environment for building and managing frontend logic and powers the server-side logic and API routing.
- npm (v8 or above): A package manager required to install dependencies for React and related libraries.
- React.js: A JavaScript library for building dynamic and responsive user interfaces.
- Browser: Google Chrome / Firefox (latest version) — For rendering and testing the UI in real-time.
- Express.js: A lightweight web framework for building RESTful APIs.
- MongoDB: A NoSQL database used to store structured and unstructured data related to users, projects, and transactions.
- Postman: Tool for testing APIs during development.
- Visual Studio Code: Preferred code editor with built-in Git and terminal support.

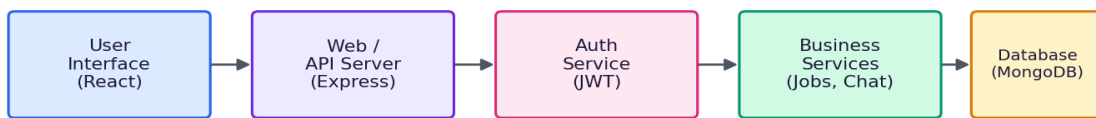
2. Hardware Requirements

Describes the minimum and recommended specifications needed to support the development and usage of the application.

- Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or better — Ensures fast compilation and multitasking during development.
- RAM: Minimum 8 GB (16 GB recommended) — For handling development servers, IDEs, and browser testing simultaneously.

- Storage: At least 1 GB free space — Required for package installations, MongoDB setup, and local project files.
- Display: 1366x768 or higher — Recommended for optimal coding experience and application layout visualization.

TECHNICAL ARCHITECTURE:



Technical Architecture Overview

FreelanceHub is built on a robust, scalable architecture designed to deliver a trustworthy and seamless hiring experience.

User Interface (UI): A clean, intuitive interface enables users to easily browse jobs, submit proposals, and manage projects. It's designed to be transparent and confidence-building, keeping both clients and freelancers focused on their goals.

Web Server: Hosts the UI and handles user requests, delivering dynamic, role-based content while ensuring smooth performance and quick response times.

API Gateway: Acts as the central entry point for all client requests, managing communication between services and ensuring efficient data flow across the system.

Authentication Service: Secures user access through JWT-based authentication and role-based authorization, protecting user data and ensuring privacy.

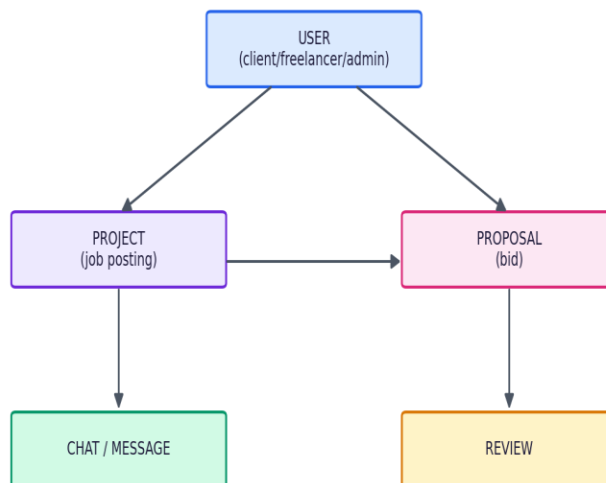
Database: Stores user profiles, job postings, proposals, and progress data, enabling personalized recommendations and reliable project tracking.

Job & Proposal Service: Manages job postings, bidding, and hiring workflows, guiding both clients and freelancers through a structured process.

Progress Tracking Service: Monitors project status, tracks milestones, and provides notifications to keep both sides informed and engaged.

Together, these components create a powerful ecosystem that supports a dynamic, personalized, and scalable freelancing experience for users.

ER DIAGRAM:



Entities and Relationships

1. User

- Represents the clients, freelancers, and admins using the platform.
- Each user has a unique profile with details such as name, email, role, and preferences.

Relationships:

- Posts → Project: A client posts one or more projects.
- Submits → Proposal: A freelancer submits proposals to projects.

2. Project

- Tracks each job posting's details, budget, required skills, and current status.
- Stores data like title, description, budget, duration, and hired freelancer.

Relationships:

- Receives → Proposal: Each project can receive multiple proposals from freelancers.

3. Proposal

- Represents an individual bid submitted by a freelancer for a project.

Relationships:

- Belongs to → Project: Proposals are tied to a specific project.
- Submitted by → User: Each proposal is linked to the freelancer who submitted it.

4. Chat / Review

- Used to enable communication and to evaluate outcomes after project completion.
- Contains messages tied to a project, or ratings and comments tied to a completed project.

Relationships:

- Linked to → Project: Chats and reviews are always tied to a specific project.
- Involves → User: Both the client and freelancer participate.

Key Features:

1. User Management

User Registration and Authentication: Allows users to register as a client or freelancer, securely log in, and authenticate their identity to access personalized dashboards.

2. Job Management

Job Listings: Displays a comprehensive catalog of available jobs with details such as title, category, budget, description, and client rating.

3. Project Progress and Trust

Progress Tracking: Tracks project status across posting, hiring, in-progress, submission, and completion, displaying status badges and history.

ROLES AND RESPONSIBILITIES:

Client:-

Registration: Clients are responsible for creating an account on FreelanceHub by providing necessary details like name, email, and password.

Profile Management: Clients have access to manipulate their profiles, allowing them to update information such as email, name, and bio.

Job Posting: Clients can post new jobs, specifying title, description, required skills, budget, and timeline.

Reviewing Proposals & Hiring: Clients can review submitted proposals, message freelancers, and hire the best fit for their project.

Providing Feedback: Clients provide ratings and reviews for freelancers after project completion.

Logout: Finally, they can logout from the FreelanceHub app.

Freelancer:-

Browsing & Bidding: Freelancers can browse open jobs, filter by skill or category, and submit proposals with a cover letter and bid amount.

Portfolio Management: Freelancers manage their skills, hourly rate, and portfolio to attract clients.

Work Delivery: Freelancers submit completed work directly through the platform for client review.

Admin:-

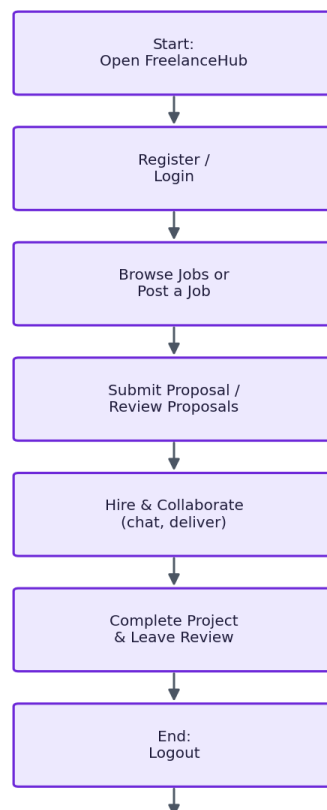
User Management: Admins are responsible for managing user accounts, including verifying and blocking accounts as needed.

Transaction & Dispute Management: Admins oversee payment transactions and resolve disputes between clients and freelancers.

Monitoring Progress: Admins monitor platform-wide activity, tracking user growth, project completion rates, and overall platform health.

Reporting and Analytics: Admins generate reports and analytics on user engagement and platform performance to make informed decisions about improvements.

User Flow:



Start: The user opens the FreelanceHub app.

Login/Register: The user selects their role — client or freelancer — and logs in or registers.

Browse or Post: Freelancers browse open jobs; clients post a new job.

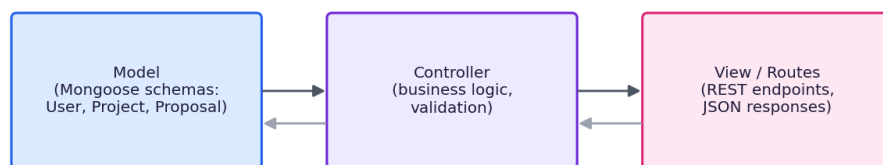
Bid or Review: Freelancers submit proposals; clients review incoming proposals.

Hire & Collaborate: The client hires a freelancer, and both sides chat and deliver work.

Complete & Review: The project is marked complete and both sides leave a review.

End: The user logs out or exits the app.

MVC PATTERN:



The FreelanceHub application follows the Model-View-Controller (MVC) architectural pattern, a software design approach that separates an application into three interconnected layers. This separation allows for modularity, easier maintenance, and scalability.

Model Layer (Data Layer)

The Model layer is responsible for handling all data-related logic. This includes the definition of data schemas (User, Project, Proposal, Chat, Review, Transaction) and the operations performed on the database using those schemas. The models are implemented using Mongoose, which provides a schema-based solution to model application data for MongoDB.

Controller Layer

The Controller layer acts as an intermediary between the view (routes) and the model. It receives incoming requests, processes the input (which may include validation or transformation), calls the appropriate methods from the model, and then returns a response to the client.

View Layer (Routing Layer)

In the context of a backend REST API, the View is implemented as the routing layer, where various endpoints are defined. These endpoints determine how the backend responds to different HTTP requests (GET, POST, PUT, DELETE) and are responsible for invoking the appropriate controller functions.

Advantages of Using MVC in This Project

- Separation of Concerns: Each layer has a clearly defined responsibility, improving readability and maintainability.
- Scalability: New features can be added easily by creating new routes, controllers, and models.
- Reusability: Logic in controllers and models can be reused across multiple parts of the application.
- Testing: Each layer can be tested independently, especially the controllers and models.
- Collaboration-Friendly: Multiple developers can work simultaneously on different layers without conflict.

Project Setup And Configuration.

Creating project folder

1. Create a new folder named `freelancehub`.
2. Inside that folder create two new folders.
3. Name one as `backend`.
4. Name another one as `frontend`.
5. Now open that folder in VS Code.

Frontend setup (installing React app)

- Open the frontend folder in the terminal of VS Code.
- Run: npm install
- To start the React server, run: npm start

Backend setup (npm init)

- Open backend folder in terminal of VS Code.
- Run: npm install
- Create files: server.js
- Create folders: models, controllers, routes, middleware

BACKEND DEVELOPMENT:

Backend/

config/

db.js — Sets up and connects your backend to the database (MongoDB Atlas).

controllers/

- projectController.js — Handles job-related logic like creating, fetching, or updating projects.
- userController.js — Handles user-related logic like signup, login, and profile operations.
- proposalController.js — Handles proposal submission, shortlisting, and withdrawal.
- adminController.js — Handles user verification, blocking, and dispute resolution.

middleware/

- auth.js — Verifies JWT tokens and protects routes that require authentication and specific roles.

models/

- Project.js — Defines the Mongoose schema/model for job postings stored in the database.
- User.js — Defines the Mongoose schema/model for user accounts.

routes/

- authRoutes.js — Defines all authentication-related API routes (login/signup).

- `projectRoutes.js` — Defines all project/job related API routes.

`server.js` — Entry point of the backend; sets up Express server, middleware, and routes.

DATABASE DEVELOPMENT:

1. Configure MongoDB:

- Install Mongoose. In your Node.js project directory, run: `npm install mongoose`
- Create database connection.
- Create Schemas & Models.

Now, make sure the database is connected before performing any of the actions through the backend, using a connection file similar to `config/db.js` described above.

Firstly, configure the Schemas for MongoDB database, to store the data in a structured pattern. Use the entities from the ER diagram above to create the schemas.

`Project.js` — Defines the Mongoose schema/model for job postings stored in the database, including title, description, budget, and status.

`User.js` — Defines the Mongoose schema/model for user accounts, including role, skills, and rating.

FRONT-END DEVELOPMENT:

Frontend/

`src/`

`components/`

- `Navbar.jsx` — Top navigation bar for the app.
- `Sidebar.jsx` — Role-aware sidebar navigation for authenticated pages.
- `Footer.jsx` — Displays the site footer.

`components/ui/`

Reusable UI components (buttons, cards, inputs, etc.).

`context/`

- AuthContext.jsx — Provides global user authentication state across the app.

pages/

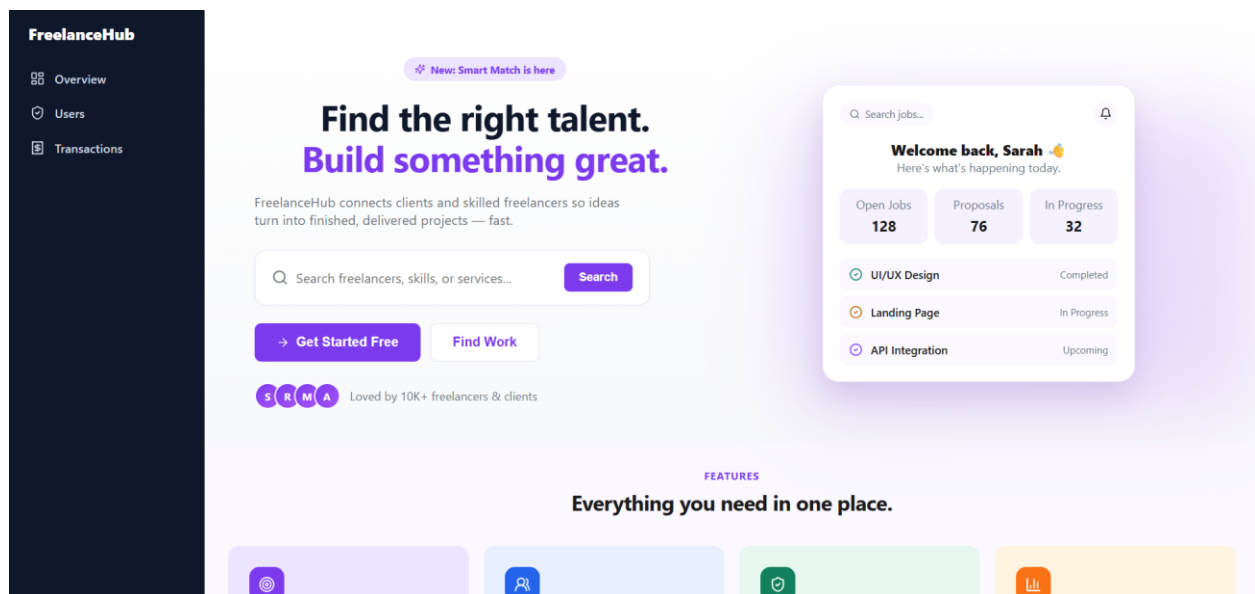
- Home.jsx — Renders the landing page UI.
- Login.jsx — Form and logic for user login.
- Register.jsx — Form and logic for new user registration.
- ProjectList.jsx — UI for browsing/filtering open jobs.
- ProjectDetail.jsx — Job details page with bidding and hiring actions.
- Dashboard.jsx — Role-based dashboard for clients and freelancers.
- Messages.jsx — Chat inbox and conversation view.

Root Files/

- App.js — Main component that holds routes and global layout.
- index.js — Entry point that renders the App component into the DOM.
- .env — Stores frontend environment variables (API URL).
- .gitignore — Files and folders Git should ignore.

Output ScreenShots:

Landing Page:



Signup Page:

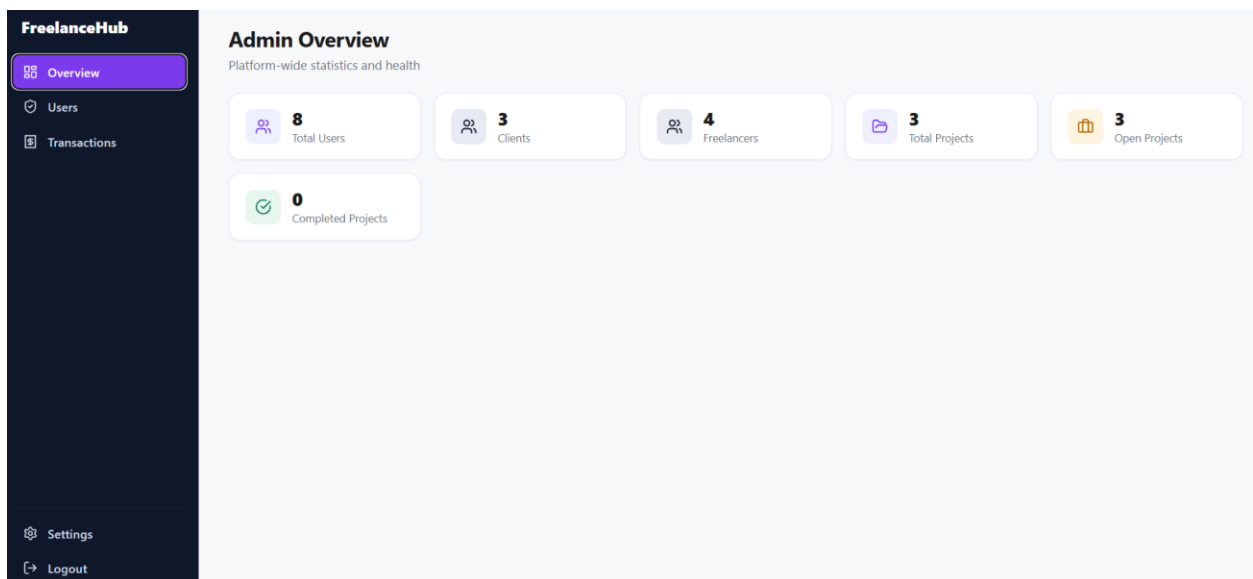
Create Account

I'm a Client (hiring) ▾

Register

Already have an account? [Log in](#)

Admin Dashboard:



FreelanceHub

Overview

Users

Transactions

Settings

Logout

Users

Verify accounts and manage access

All

Clients

Freelancers

Admins

Name	Email	Role	Verified	Status	Actions
deepa	di@exx.com	freelancer	—	Active	Verify Block
siya	ex2@ex.com	client	—	Active	Verify Block
riya	ri@example.com	freelancer	—	Active	Verify Block
diya	vb@ex.com	client	—	Active	Verify Block
Patrisha	example1@example.com	admin	—	Active	Verify Block
Arya Sakore	example@gmail.com	freelancer	—	Active	Verify Block
Samruddhi Sanjay Yawale	samyawale2005@gmail.com	client	—	Active	Verify Block
Sarah	sarah@test.com	freelancer	—	Active	Verify Block

FreelanceHub

Overview

Users

Transactions

Transactions & Disputes

Review payments and resolve disputes

All

Pending

Held

Released

Refunded

Disputed

Client Dashboard:

FreelanceHub

Dashboard

Post a Job

Find Freelancers

My Jobs

Messages

Profile

Settings

Logout

Welcome back, vaish 🍷

Here's what's happening with your projects

Post a Job

0

Active Jobs

0

In Progress

0

Total Projects

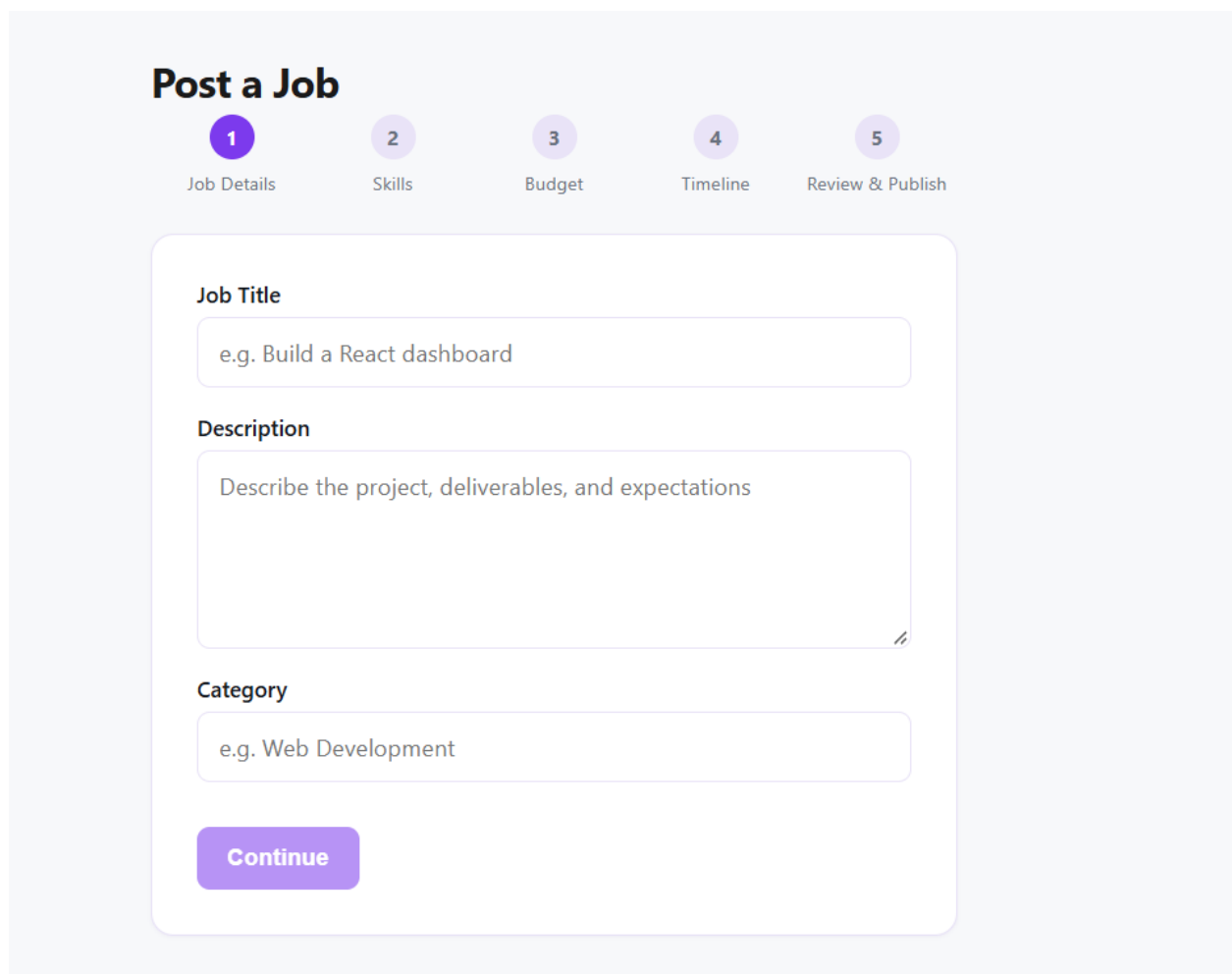
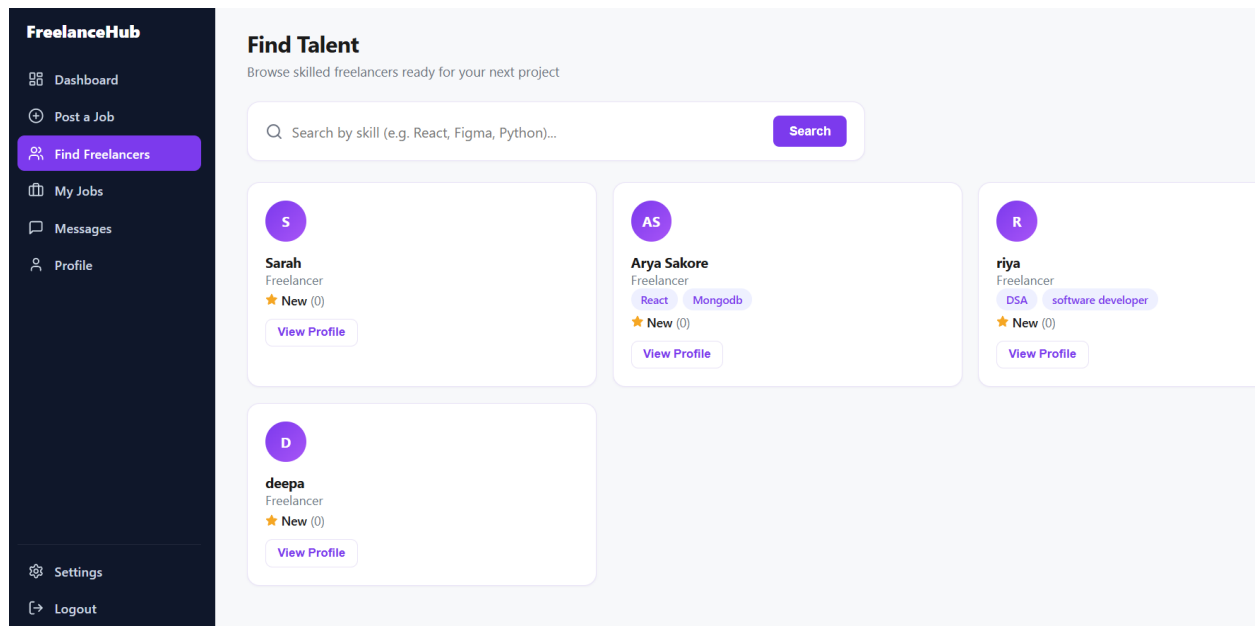
\$0

Total Spend

My Projects

No projects posted yet

Post your first job to start hiring freelancers



Freelancer Dashboard:

FreelanceHub

Dashboard

Find Jobs

Proposals

Messages

Earnings

Profile

Settings

Logout

Good morning, divya 🍌

Here's an overview of your freelance activity

\$0

Total Earnings

0

Active Projects

0

Pending Proposals

New

Average Rating

Find Jobs

My Proposals

No proposals submitted yet

Browse open jobs and submit your first proposal

Find Jobs Page:

FreelanceHub

Dashboard

Find Jobs

Proposals

Messages

Earnings

Profile

Settings

Logout

Find Jobs

Browse open projects that match your skills

Search jobs, skills, or keywords...

Search

Category

Web Development

AI / ML

Data Science

UI/UX

Mobile Development

Digital Marketing

Budget

\$0 – \$500

\$500 – \$2,000

\$2,000 – \$5,000

\$5,000+

Apply Filters

e commerce app

8h ago

siya

business application

react DSA

\$8900

8-9weeks

View Job

Apply Now

E-commerce

8h ago

diya

shopping app

React MongoDB

\$8500

5-6 weeks

View Job

Apply Now

Build a website for healthcare

23h ago

Samruddhi Sanjay Yawale

bchygefvgbrjfcvjvb.bj vijfcb huvhbcvbfhdvvhdbhbwh

react mongodb etc

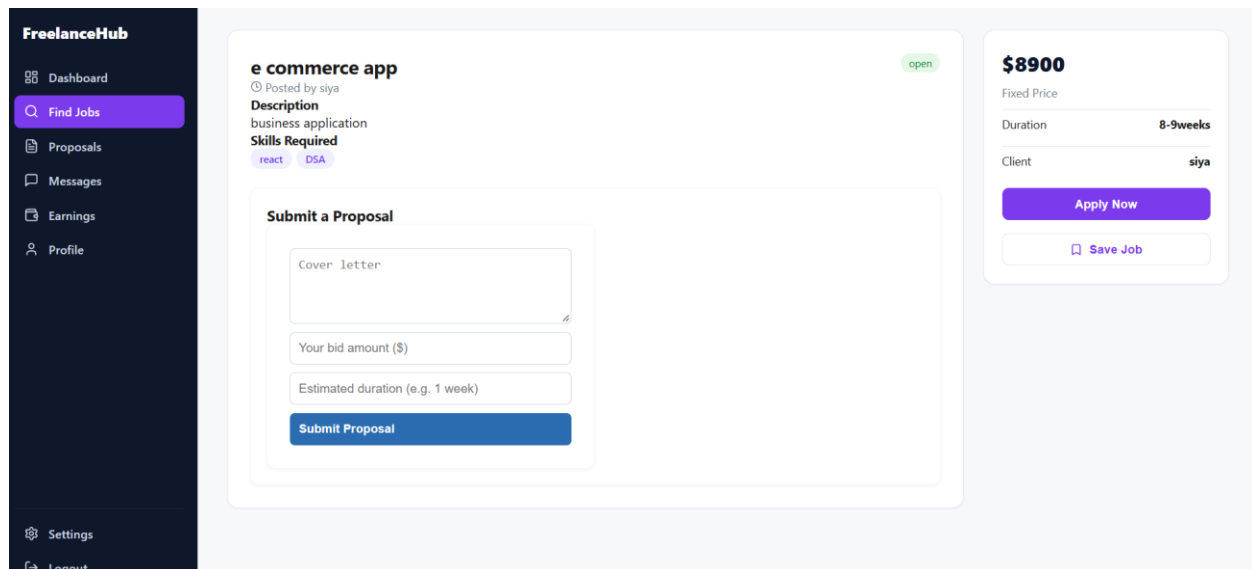
\$6999

6 weeks

View Job

Apply Now

Job Details Page:



Demo Video Link: "https://indiraicemac-my.sharepoint.com/:v/g/personal/samruddhi_yawale_indiraicem_ac_in/IQBN1FQVsFocQpYVE1x2kS1aATo7gkZV2fefBuYX4W2VjaY?nav=eyJyZWZlcnJhbEluZm8iOmsicmVmZXJyYWxBcHAIoiJPbmVEcm12ZUZvckJlc2luZXNzliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkiX0&e=Uoh2Fn"

Code Drive Link: <https://github.com/samruddhi01-01/Freelancehub>